

Random Variables

- Random Variable: Consider a random experiment with a sample space $\mathcal{C}$. A function X , which assigns to each element $c \in \mathcal{C}$ one and only one number $X(c) = x$, is called a **random variable**. The **space** or **range** of X is the set of real numbers $\mathcal{D} = \{x : x = X(c), c \in \mathcal{C}\}$.
- For example: Toss a coin, $\mathcal{C} = \{H, T\}$; $\mathcal{D} = \{0, 1\}$
- Cumulative distribution function (cdf): $F(x) = \mathbb{P}(X \leq x)$
- Discrete random variables: $\mathcal{D}$ is a countable set
 - Probability mass function (pmf): $p(x) = \mathbb{P}(X = x)$
- Continuous random variables: $\mathcal{D}$ is an interval of real numbers
 - Probability density function (pdf): $f(x) = \frac{d}{dx} F(x)$

- Point mass: $X \sim \delta_c$
- Discrete Uniform: $X \sim \text{unif}\{c_1, \dots, c_k\}$
- Bernoulli: $X \sim \text{Bernoulli}(p)$

Some Well-known Discrete Random Variables

- Binomial: $X \sim \text{Bin}(n, p)$
- Geometric: $X \sim \text{Geometric}(p)$
- Poisson: $X \sim \text{Poisson}(\lambda)$

Some Well-known Continuous Random Variables

- Uniform: $X \sim \text{unif}[a, b]$
- Exponential: $X \sim \text{exp}(\beta)$
- Normal: $X \sim N(\mu, \sigma^2)$